WEEKLY PROGRESS REPORT

Mechatronic System Design (MEC483) — Course Project

PROJECT / PHASE

Smart Pick and Place Robot

DATE RANGE

24/3/2026 – 31/3/2026

TEAM MEMBERS

Maher Abo Abed (1087993) , Sabeeha Zainab Hasham (1089042), Basel Feras Ghunaim (1088912), Ahmad Nasser Alshehhi (1090882)

HOW ARE WE DOING THIS WEEK?

TOP WINS (MAX 3)

- Started the conceptual design phase, which includes design requirement, constraints ,sketches, and CAD models for the different systems (Platform, rotary joint, and prismatic joint)
- Building a foundation for the electrical architecture (battery requirements, motor drivers, step down, cables, power unit)
- Used the HSV colour detection system alongside shape detection using Opencv to identify an object within a test space.

KEY BLOCKERS / RISKS (MAX 3)

- Blocker : Limited university access, making gathering as a group extremely difficult.
- Risk: The components are pricey, so every step has to be studied carefully before executing.
- Blocker: Uncertainty around the future of the semester is creating hesitation and doubt.

PROJECT HEALTH

Scope	Under Execution
Schedule	Potential Delays
Fabrication & Prototyping	On Halt

OPEN ISSUES

3 Active

Project Schedule (Gantt)

Agile Gantt chart1.xlsx

MILESTONES & DELIVERABLES

Milestone / Deliverable	Due Date	Owner	Status / Evidence
Sensor System Definition (Week 1 Report)	–	Member 3 & Member 4	COMPLETED / PDF Documentation
Motor Selection & Analysis (PCC Method)	–	Member 2	COMPLETED / Decision Matrix
Encoder & Position Strategy Evaluation	–	Member 1 & Member 3	COMPLETED / Technical Review
Vacuum Suction System Design	–	Member 3	COMPLETED / Force Calculation
Pump Selection Finalization	–	Member 2	IN PROGRESS
Hybrid End Effector Concept (Gripper + Suction)	–	Member 2 & Member 3	IN PROGRESS
Sensor Integration & System Architecture	–	Member 4 & Team	IN PROGRESS
Motion Control Implementation (Stepper Control Code)	–	Member 3	PENDING
CAD Model of Prototype	–	Member 1 & Team	IN PROGRESS
Mechanical Prototype Assembly	–	Team	NOT STARTED
System Integration Testing	–	Team	NOT STARTED
Final Validation & Demonstration	–	All Members	NOT STARTED

ISSUES, RISKS, AND ACTION ITEMS

TOP ISSUES (BLOCKING PROGRESS)

- Being limited to solely software-based work, ie: research/computer vision can slow down performance – not having easy access to facilities to make the model, ie: 3D printer.
- Due to current circumstances, teammates cannot meet in person to discuss project planning, this has resulted in Microsoft Teams meetings occurring 1-2 times a week. This can be limiting as only one teammate has access to the objects that need to be picked up by the arm.

MITIGATIONS / DECISIONS NEEDED

- Focus on aspects that can be conducted without the need for printing the model – design/computer vision/research
- More comprehensive meetings are required to mitigate this issue, planning in advance and strong delegation of tasks needs to be implemented.

ACTION ITEMS (NEXT 7 DAYS)

ID	ACTION ITEM DESCRIPTION	MEMBER	WEEK DUE
A1	Investigate how to mitigate issues experienced using vision detection	Sabeeha	Week 4
A2	Finalise the main CAD design and what is required in order to make the prototype.	Maher	Week 4
A3	Investigate motion control strategy, torque estimations, suction system and gripper	Basel	Week 4
A4	Finalise sensor selection and integration in system	Ahmad	Week 4

INDIVIDUAL UPDATES

WEEKLY PROGRESS & NEXT STEPS

MEMBER 1

Maher Abo Abed

MEMBER 2

Sabeeha Zainab Hasham

MEMBER 3

Basel Feras Ghunaim

MEMBER 4

Ahmad Nasser Alshehhi

MEMBER 1: Maher Abo Abed

ACCOMPLISHED (SINCE LAST REPORT)

- Design definition (Joints, system architecture, environment planning, operation order)
- System complexity reduction (Smart container placement,)
- Started looking into the electrical architecture (Battery size, step down, motor drivers (A4989, PCA9685)
- Robots platform requirements (what components should there be space for?)
- Supporting components (Thrust, radial, ball bearing, flanges)
- Joint mechanisms (lead screw, GT2 pulleys)
- Explored existing designs
- Sketches

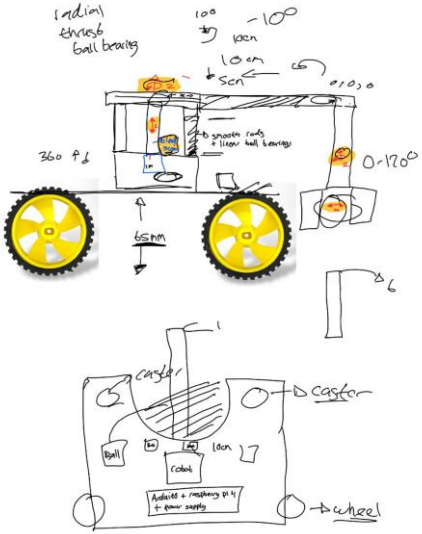
IN PROGRESS

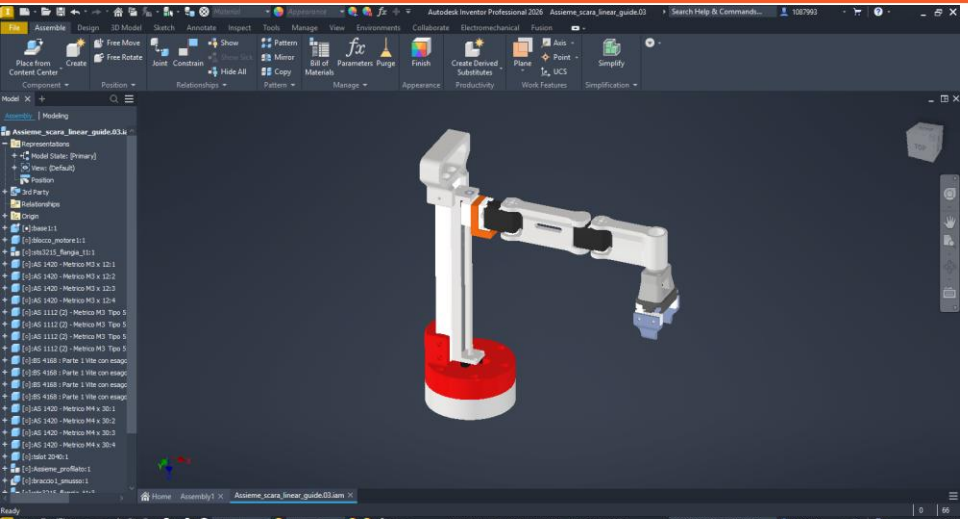
- CAD Design
- Exploring Open-Source frameworks
- Research
- Electrical Architecture

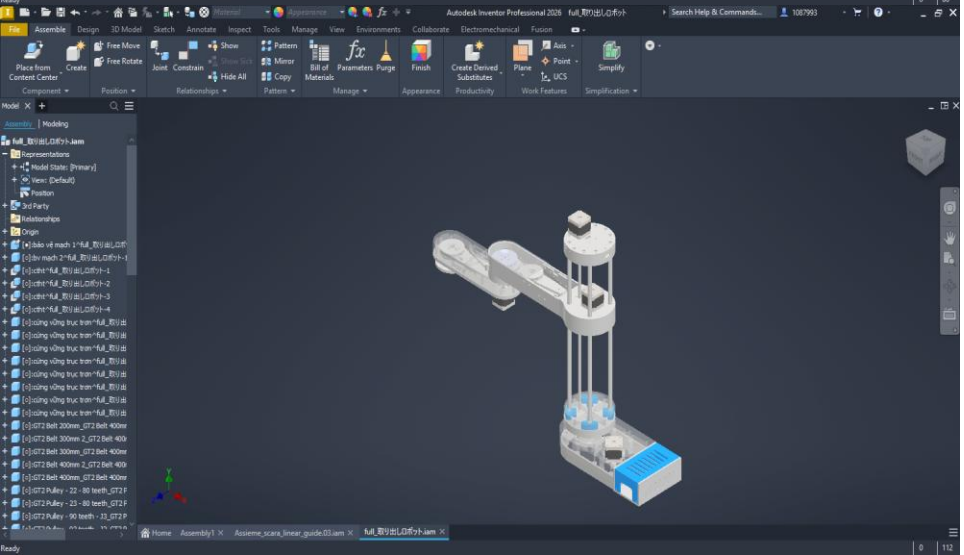
NEXT WEEK PLAN

- Present a CAD design
- Start fabrication
- Procurement of components

EVIDENCE







Joint 1: Base joint (0-360°), rotation

- Stepper motor + maybe small gear system
- Bearings
- Micro limit switch

Joint 2: Vertical, linear

- Stepper motor
- Lead screw
- smooth rods
- linear ball bearing
- ball bearings
- Shaft coupler
- micro limit switch

Joint 3: Elbow (0-120°), rotation

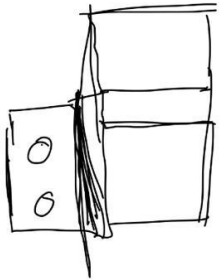
- Servo
- Connecting parts/fittings

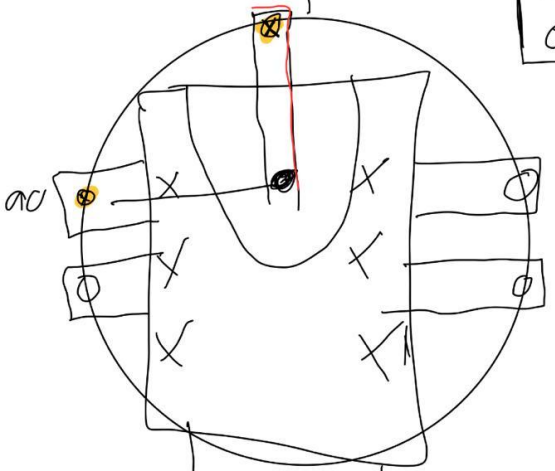
Join 4: Wrist (0-120°), rotation

- Servo
- Connecting parts/fittings

Platform:

- Base of the robot
- Boxes coming out
- Arduino R3
- Raspberry pi 4
- Battery 12V
- Breadboard
- Motor drive (HW288/PCA9685)
- Step down converter
- Camera (subject to change)





MEMBER 2: Sabeeha Hasham

ACCOMPLISHED (SINCE LAST REPORT)

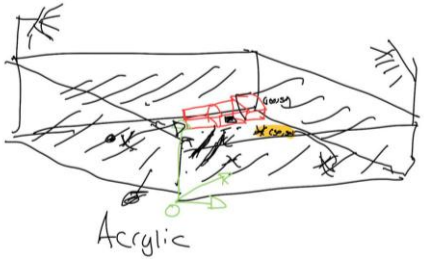
- Used opencv to implement colour detection to convert images from BGR to HSV and understanding the HSV colour space and how it can be further used.
- Adjusted the threshold values to add ranges making it easier to isolate the object from its background, whilst testing shape detection as well.
- Discussed CAD designs and how the setup will be implemented, the environment and layout of the project

IN PROGRESS

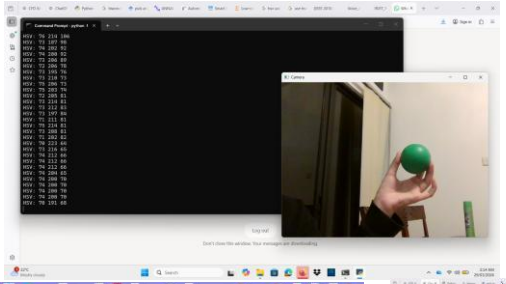
- Trying to implement colour or shape detection on the bolt and nut, some obstacles have been run into due to the colour ranges of a metal being quite large and visual noise because of the background makes it hard to identify the object – it is important to take into consideration changing angles when in a pick and place robot.

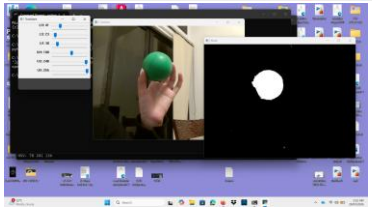
NEXT WEEK PLAN

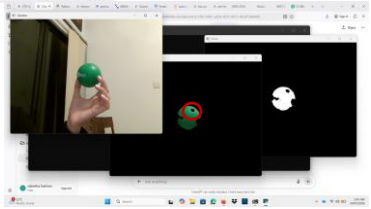
- CAD design and shape detection refining.

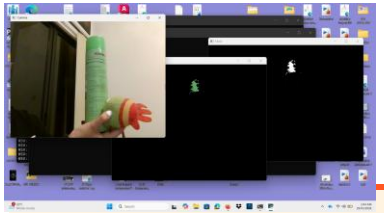


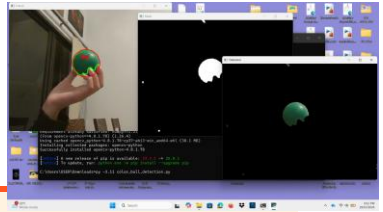
EVIDENCE (SHOW, DON'T TELL)











HELP NEEDED / QUESTIONS

- Since the robot is mobile, can we operate it in its own closure?

MEMBER 3: Basel Feras Ghunaim

ACCOMPLISHED (SINCE LAST REPORT)

- Completed evaluation of motion control strategies (AccelStepper vs custom step generation).
- Finalized decision to use stepper motor step-counting with homing as the primary positioning method.
- Conducted detailed analysis of encoder integration options (single encoder, dual encoder, and comparison with motor step tracking).
- Refined hybrid end-effector concept (mechanical gripper + suction cup) and validated feasibility for all target objects.
- Investigated vacuum pump specifications and confirmed suitability of electric vacuum pump over compressed air systems.

IN PROGRESS

- Implementing initial stepper motor control logic and testing basic motion behavior.
- Evaluating potential positioning errors due to gear backlash and belt elasticity.
- Optimizing suction cup placement and integration within the gripper design.
- Reviewing control sequence timing for coordinated pick-and-place operations.

MEMBER 4: Ahmed Alshehhi

ACCOMPLISHED (SINCE LAST REPORT)

- Reviewed research papers and technical resources on robotic gripper design.
- Analyzed different gripper types (mechanical, soft grippers, and vacuum-based systems).

IN PROGRESS

- Selecting the most suitable gripper design based on object shape, weight, and fragility
- Studying methods to control gripping force to ensure safe handling of delicate objects (e.g., egg)

NEXT WEEK PLAN

- Finalize the gripper design and add it to the CAD model
- Begin testing basic gripping concepts using a simple prototype

HELP NEEDED / QUESTIONS

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WRAP-UP: WHAT WE NEED FROM THE INSTRUCTOR / STAKEHOLDERS

DECISIONS REQUESTED

- Decide and work on the final the design.
- Operating procedure
- Electrical architecture
- Software
- Computer vision model strategy

QUESTIONS / FEEDBACK

- Can we create our own testing roles and conditions to suit our design?
- Any updates with regards to coming back to the university?

ACTION ITEMS FOLLOW-UP

- Begin initial CAD modeling and system integration
- Refine and finalise a vision detection and classification system to identify the objects
- Make final decisions on the workspace that the arm will operate in, everything related to size of workspace/colour, etc.

APPENDIX

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